

Image Registration – 1

References:

1. J. West, *et. al.*, Comparison and evaluation of retrospective intermodality image registration techniques, SPIE Vol. 2710.
2. G.P. Penney, *et. al.*, A comparison of similarity measures for use in 2D-3D medical image registration, IEEE Transactions on Medical Imaging, Aug. 1998.
3. Numerical recipes in C (Chapter 10), William H. Press.
4. J.S. DeBonet and A. Chao, Structure-driven SAR image registration, Proceedings of SPIE, 1998.
5. Medical Image Registration, Hajnal, Hill and Hawkes, CRC 2001.
6. Optimization: Foundations and Applications, Ronald Miller.
7. Richard Szeliski, Image Alignment and Stitching, Technical Report, MSR-TR-2004-92, 2006.
8. William M. Wells III, *et. al.*, Multi-modal volume registration by maximization of mutual information, Medical Image Analysis, 1(1), pp. 35-51, 1996.

Example: input images



Aerial Photo Image taken from an elevated position

Floating image or Moving image



Orthophoto Image geometrically corrected

Fixed image

Example: registered images



Registered Image

Floating image or Moving image



Orthophoto Image

Fixed image

Cylindrical Panorama



(a)



(b)

Figure 7: *An example of a cylindrical panorama: (a) two cylindrically warped images related by a horizontal translation; (b) part of a cylindrical panorama composited from a sequence of images.*

Richard Szeliski

Panoramic Image Mosaic

Chaman Singh Verma and Mon-Ju



http://pages.cs.wisc.edu/~csverma/CS766_09/ImageMosaic/imagemosaic.html

Spherical Panorama



Figure 8: *An example of a spherical panorama constructed from 54 photographs.*

Registering multiple images

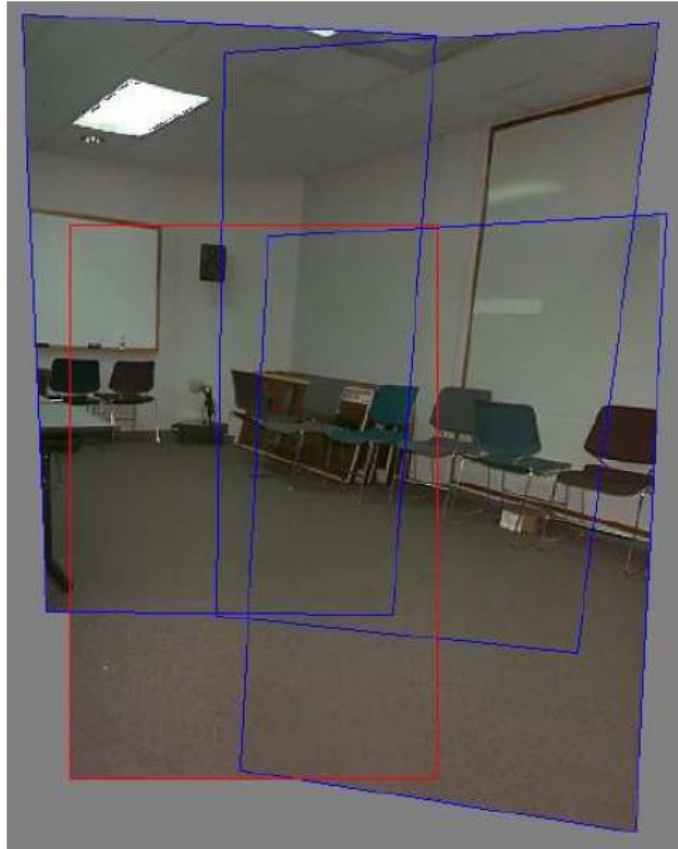


Figure 16: *Four images taken with a hand-held model registered using a 3D rotation motion model (from (Szeliski and Shum 1997)). Notice how the homographies, rather than being arbitrary, have a well defined keystone shape whose width increases away from the origin.*

Registering multiple images



Registering multiple faces

Group-wise image registration

Pair-wise image registration

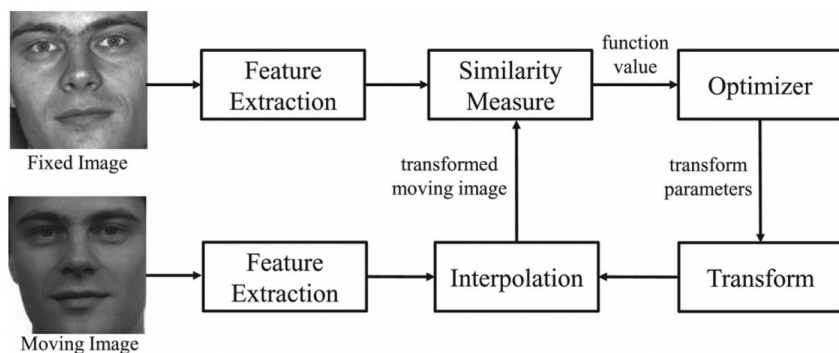


Fig. 1. The schematic illustration of the pairwise registration process.

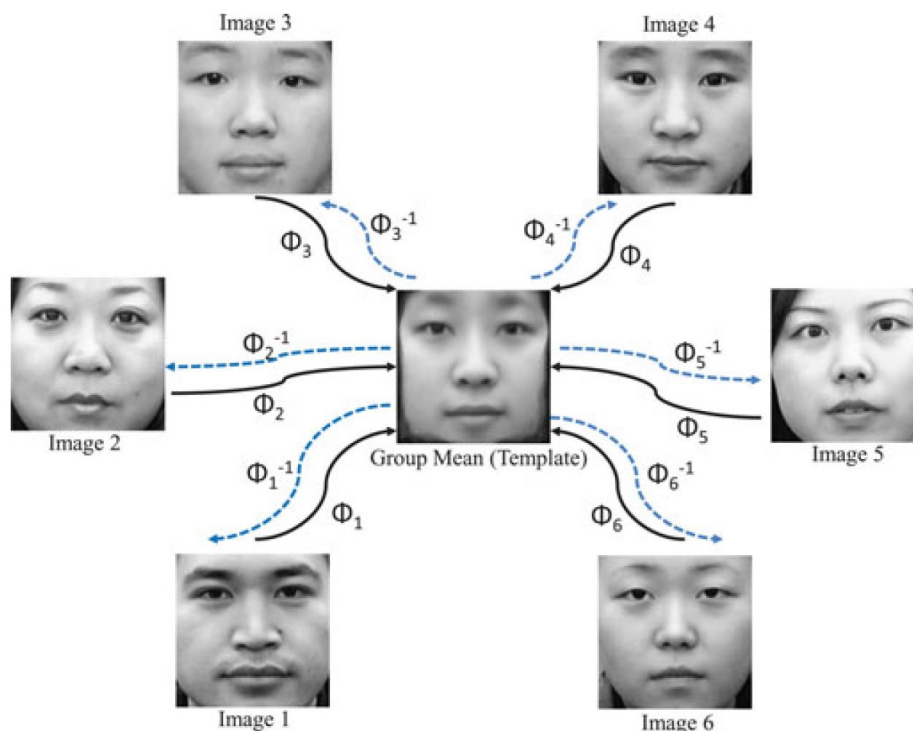
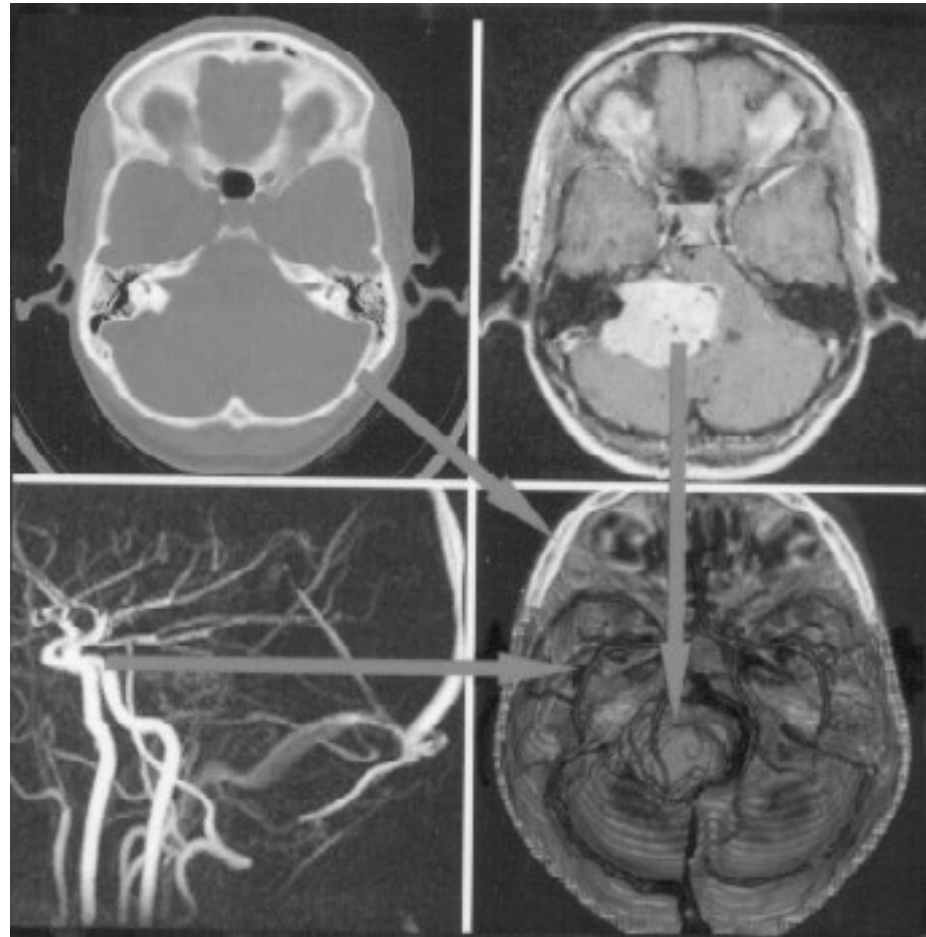


Fig. 2. The schematic illustration of groupwise registration. The template (group mean) is estimated which has the smallest geodesic distances among the six input images on the Riemannian manifold. Each image I_i can be transformed to the template space with transformation ϕ_i (black solid arrows). The template can also be warped to each individual image's space by the backward transformation ϕ_i^{-1} (blue dashed arrows).

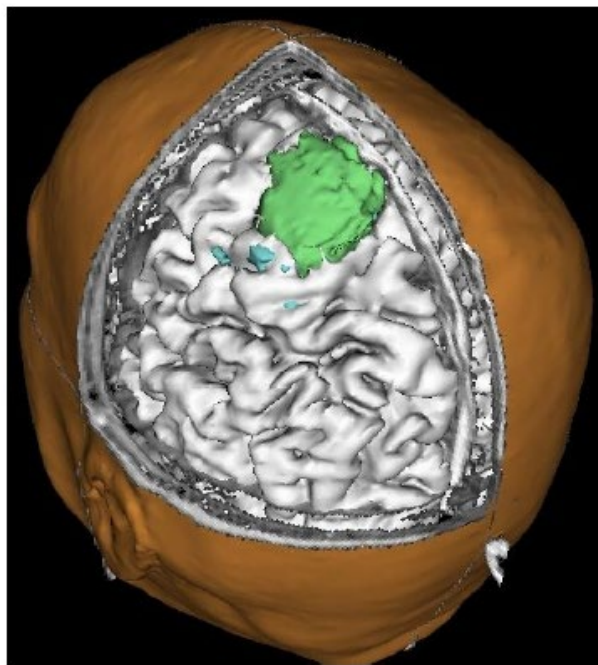
Applications in Medical Imaging

1. Multi-modality image fusion
2. Time series image processing
 - a. fMRI images
 - b. Cardiac ultrasound images
3. Image warping across patients, construction of image atlas for biomedical research
4. Accommodate tissue deformations in image-guided surgery

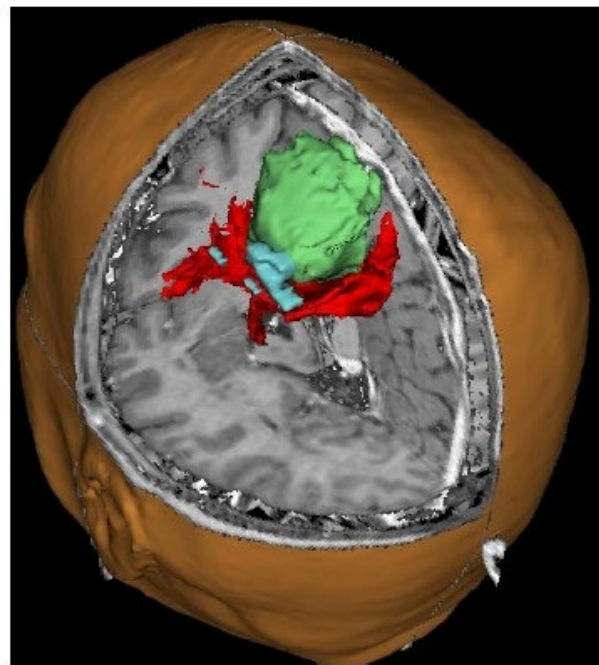


3D Slicer is a multi-platform, free open source software (FOSS) for visualization and image computing. The navigation box on the left leads to additional pages with more information.

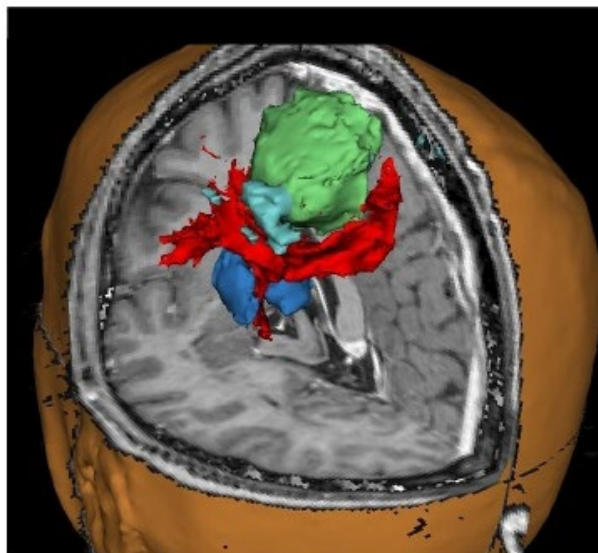




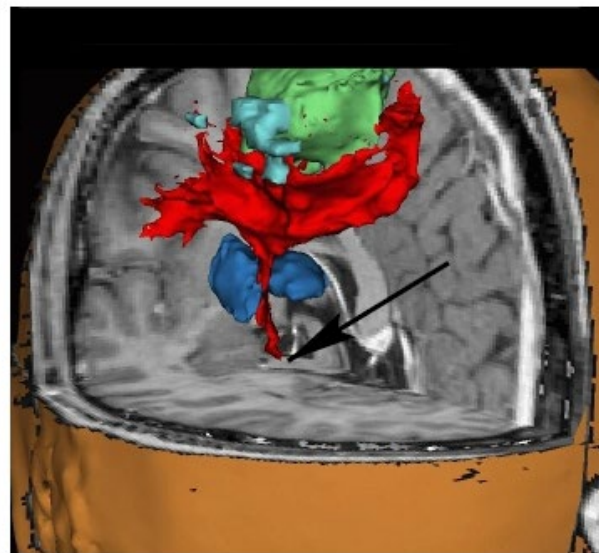
(a) Brain (White) and Tumor (Green) from MRI



(b) Corticospinal Tract (Red) and Motor Cortex from fMRI (Light Blue)



(c) Basal Ganglia (Dark Blue)



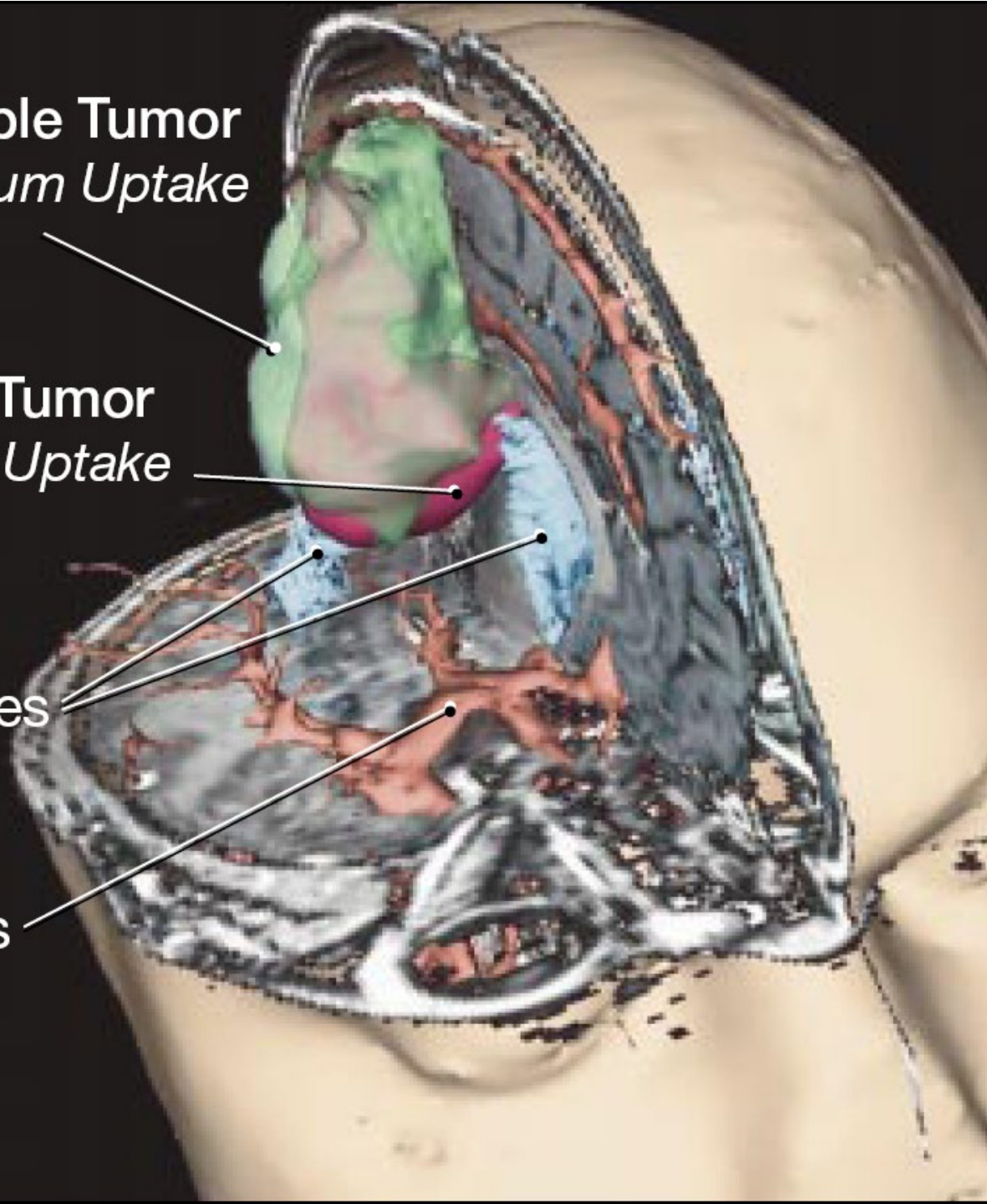
(d) Crus Cerebri (Arrow)

Nonviable Tumor
Gadolinium Uptake

Viable Tumor
Thallium Uptake

Ventricles

**Blood
Vessels**

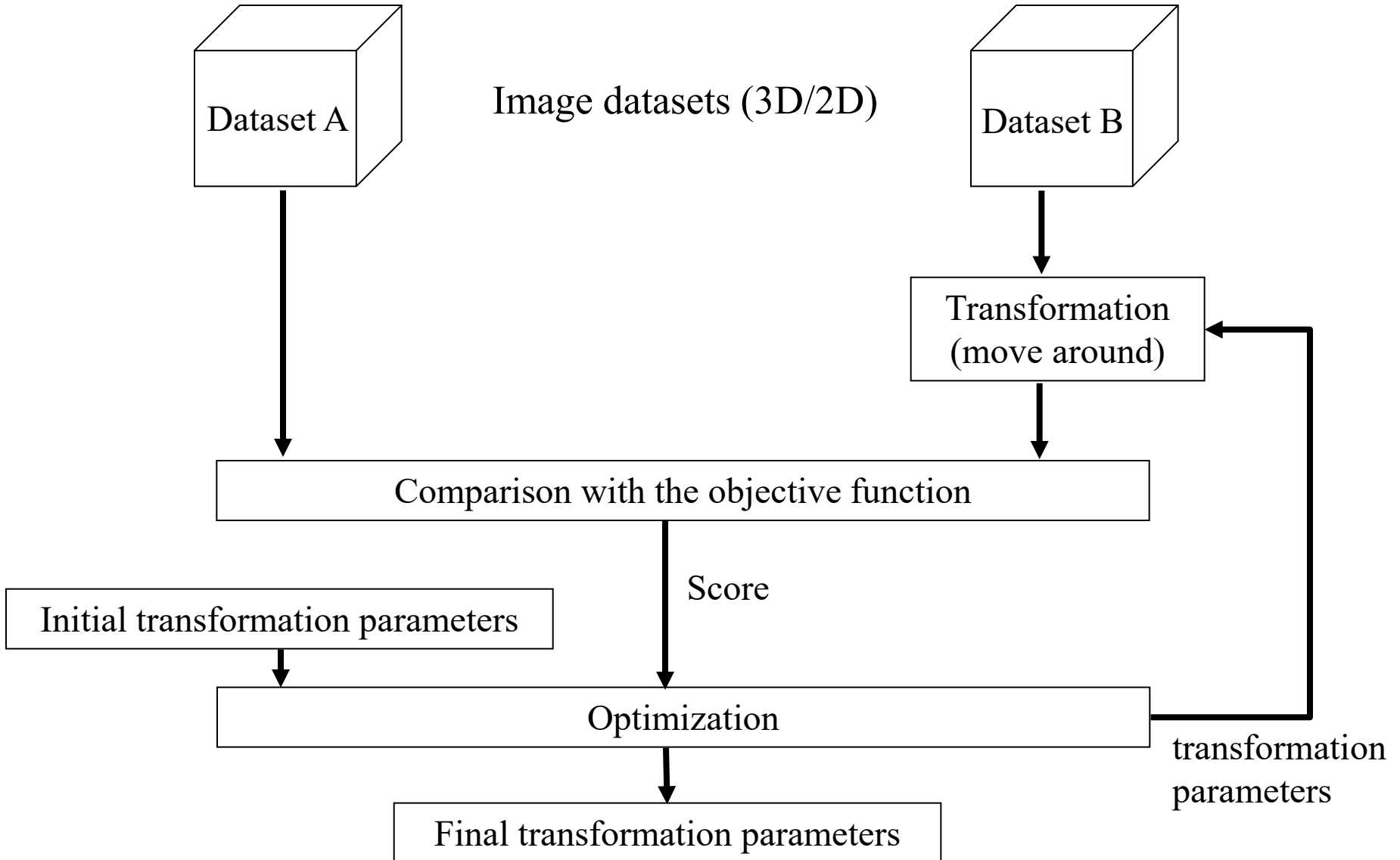


Registering multiple images

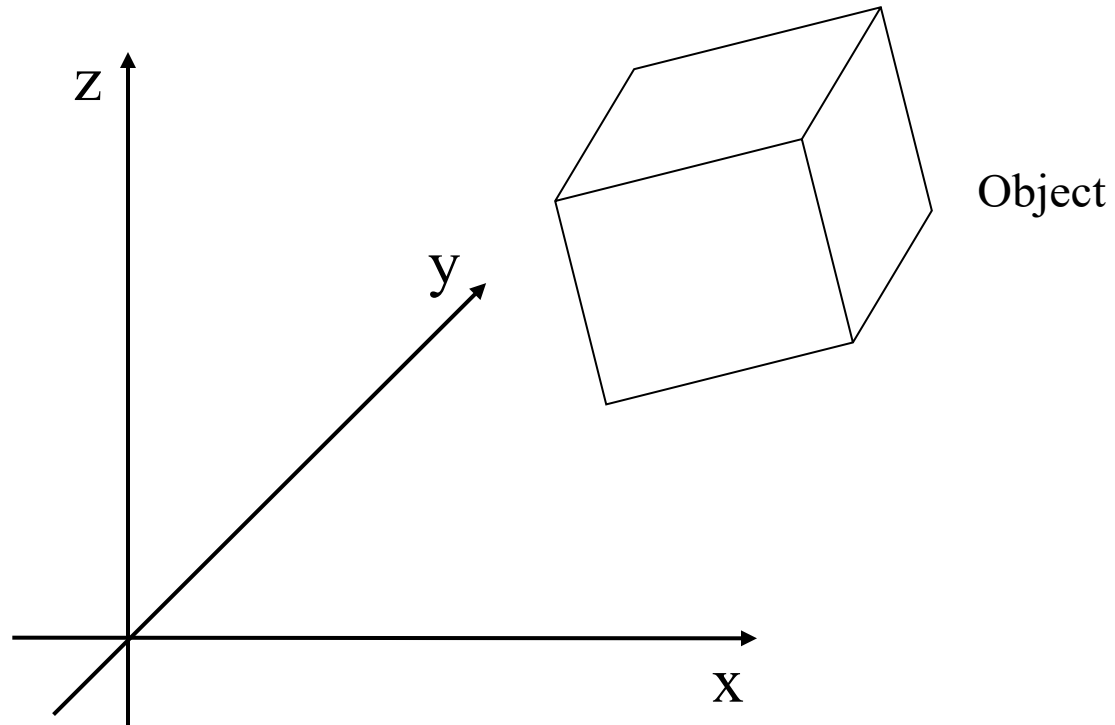


Figure 23: *From a set of five source images (of which four are shown on the left), Photomontage quickly creates a composite family portrait in which everyone is smiling and looking at the camera (right). Users simply flip through the stack and coarsely draw strokes using the designated source image objective over the people they wish to add to the composite. The user-applied strokes and computed regions are color-coded by the borders of the source images on the left (middle). (Copied, with permission, from (Agarwala et al. 2004)).*

Image registration



Object as function

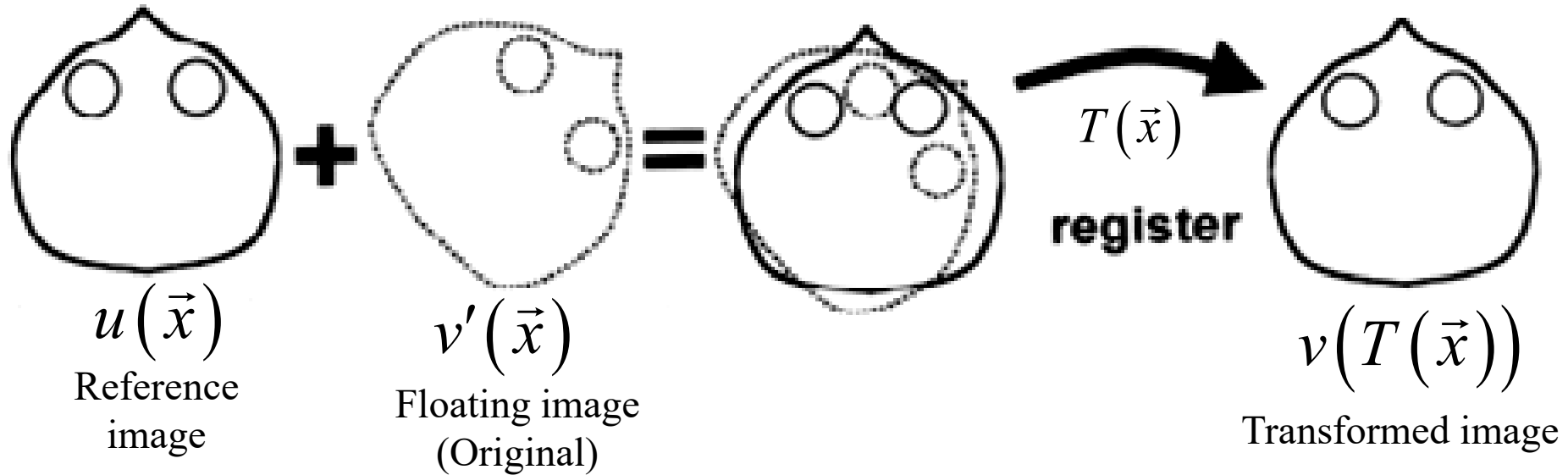


Example

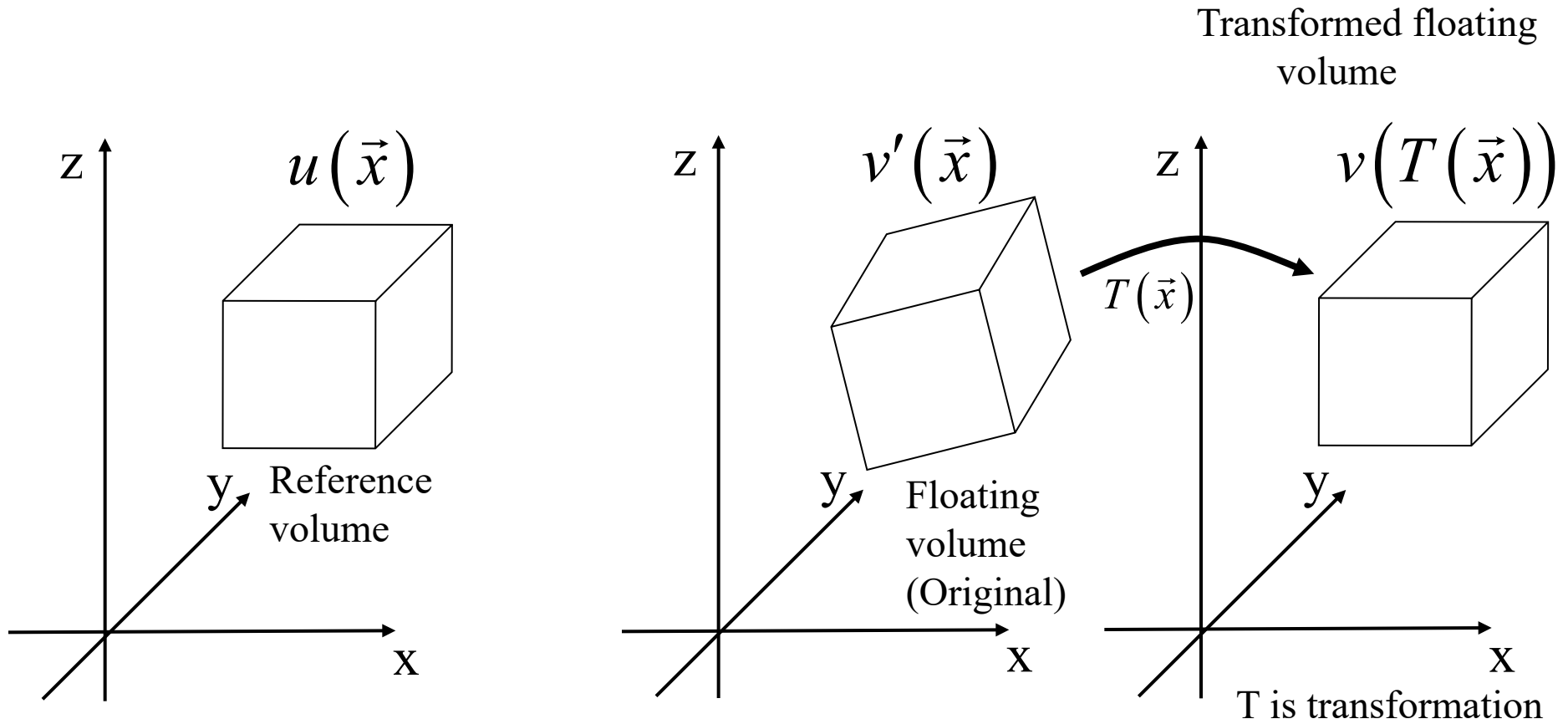
$v(\vec{x})$: coordinates $\rightarrow$ intensity value

$\vec{x} = (x, y, z)^T$, T represents transpose

Register 2D images



Register 3D volumes



Goal of image registration

$$\begin{array}{ccc} u(\vec{x}) & \Leftrightarrow & v(T(\vec{x})) \\ \uparrow & & \uparrow \\ \text{Reference} & & \text{Transformed floating} \\ \text{image/volume} & & \text{image/volume} \end{array}$$

3D Registration process:

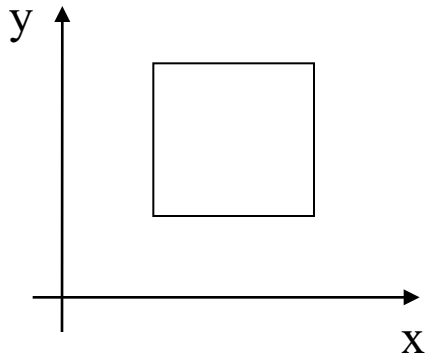
- Comparison between the *reference* and the *transformed floating* volumes based on the objective function $F(u, v(T))$ and the current transformation T .
- The goal is to find the optimal value of the objective function and the optimal transformation $\hat{T}$.

3D Registration problem can be viewed as an optimization problem:
Search for $T = (\theta_x, \theta_y, \theta_z, d_x, d_y, d_z)$ that maximizes/minimizes $F(u, v(T))$

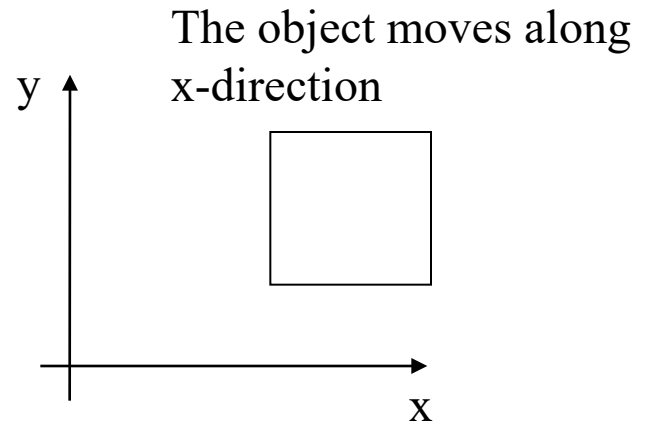
Translation

Translation $T(\vec{x}) \equiv \vec{x} + \vec{d}$

Example: 2D $T\begin{pmatrix} x \\ y \end{pmatrix} \equiv \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} d_x \\ d_y \end{pmatrix}$



$$\begin{aligned} d_x &> 0 \\ d_y &= 0 \end{aligned}$$



$$\begin{aligned} d_x &= 0 \\ d_y &\neq 0 \end{aligned}$$

The object moves along y-direction

Rotation by θ about the z axis. (Rotation is measured clockwise about the z axis while looking along the positive axis direction)

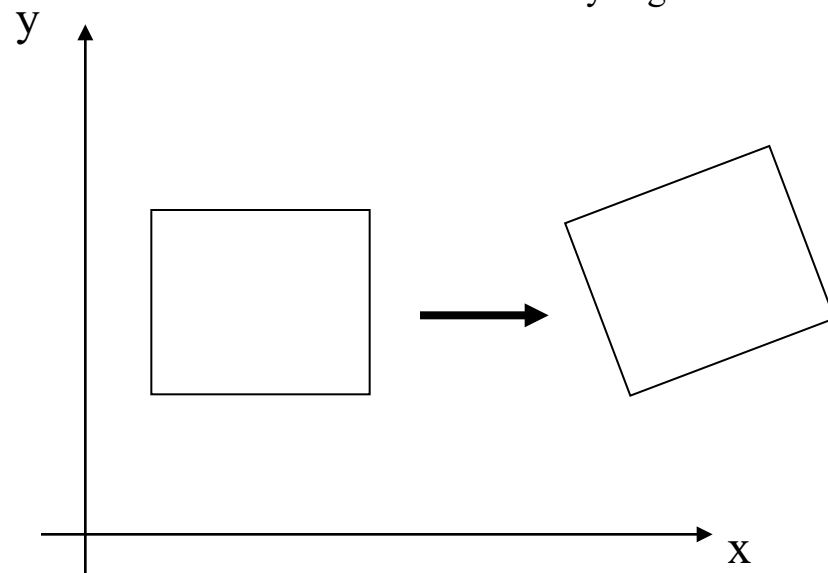
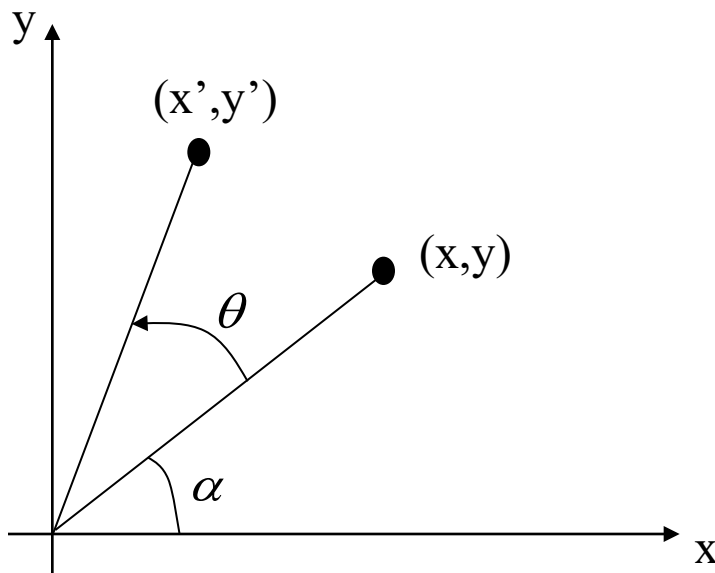
$$T(\vec{x}) \equiv R_\theta \vec{x}$$

$$R_\theta \equiv \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_\theta R_\theta^T = I$$

$$|R_\theta| = 1$$

Proved by trigonometric identities



Rotation by θ about the x axis. (Rotation is measured clockwise about the x axis while looking along the positive axis direction)

$$R_{\theta} \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$$

Proved by trigonometric identities

Rotation by θ about the y axis. (Rotation is measured clockwise about the y axis while looking along the positive axis direction)

$$R_{\theta} \equiv \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

Proved by trigonometric identities

Rigid body transformation: translation + rotation

Translation + rotation $T(\vec{x}) \equiv R\vec{x} + \vec{d}$, where $R = R_{\theta_x} R_{\theta_y} R_{\theta_z}$
Composition of rotation

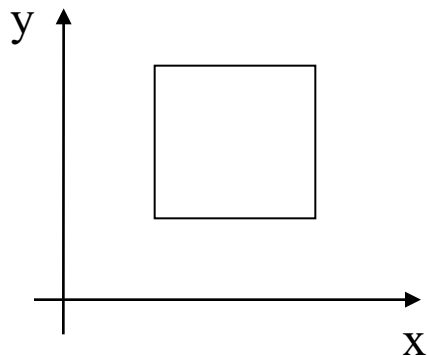
Example: 3D $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} \equiv R\begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} d_x \\ d_y \\ d_z \end{pmatrix}$ 6 degrees of freedom (DOF)

$$R = \begin{pmatrix} \cos\theta_y \cos\theta_z & -\cos\theta_y \sin\theta_z & \sin\theta_y \\ \sin\theta_x \sin\theta_y \cos\theta_z + \cos\theta_x \sin\theta_z & -\sin\theta_x \sin\theta_y \sin\theta_z + \cos\theta_x \cos\theta_z & -\sin\theta_x \cos\theta_y \\ -\cos\theta_x \sin\theta_y \cos\theta_z + \sin\theta_x \sin\theta_z & \cos\theta_x \sin\theta_y \sin\theta_z + \sin\theta_x \cos\theta_z & \cos\theta_x \cos\theta_y \end{pmatrix}$$

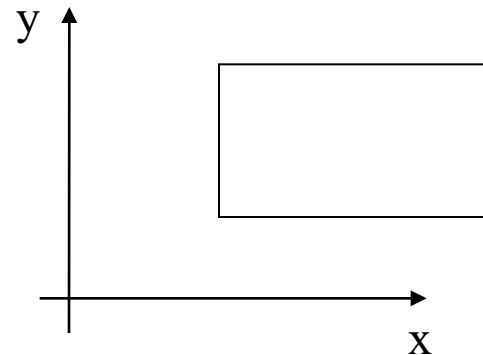
Scaling + translation

Translation $T(\vec{x}) \equiv S \vec{x} + \vec{d}$

Example: 3D $T \begin{pmatrix} x \\ y \\ z \end{pmatrix} \equiv \begin{pmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & S_z \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} d_x \\ d_y \\ d_z \end{pmatrix}$



$$S_x \neq 1$$
$$S_y, S_z = 1$$

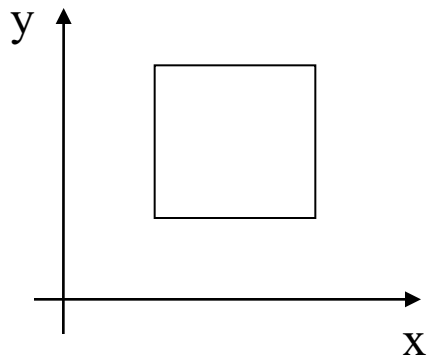


The object is stretched along the x-direction

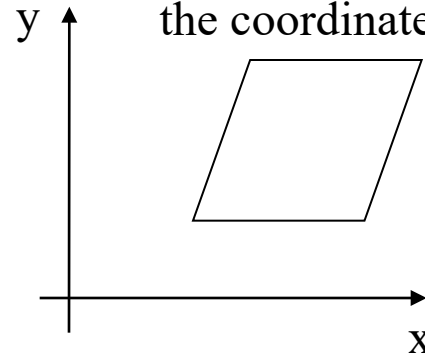
Shearing + translation

Translation $T(\vec{x}) \equiv K \vec{x} + \vec{d}$

Example: 3D $T \begin{pmatrix} x \\ y \\ z \end{pmatrix} \equiv \begin{pmatrix} 1 & k_{xy} & k_{xz} \\ k_{yx} & 1 & k_{yz} \\ k_{zx} & k_{zy} & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} d_x \\ d_y \\ d_z \end{pmatrix}$



$$k_{xy} \neq 0$$
$$\text{Other } k = 0$$



The object is linearly stretched along the x-direction based on the coordinates of each point.

Affine transformation for 3D non-rigid registration

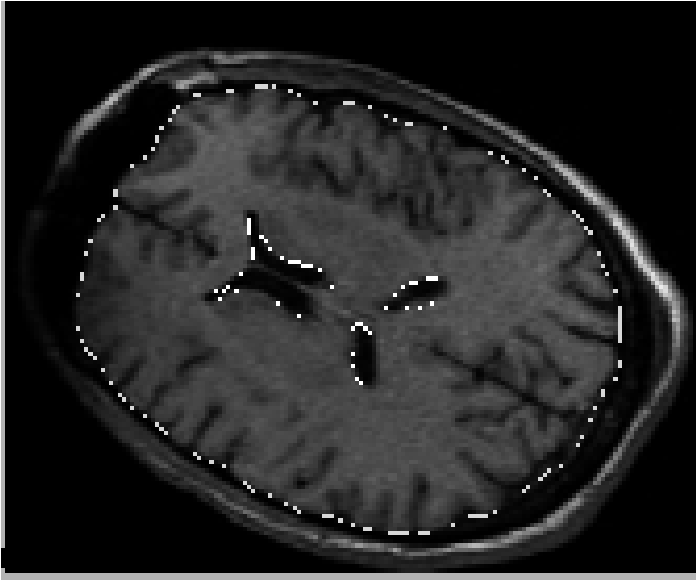
1. Shearing and scaling are included.
2. Beyond rigid body motion (translations and rotations only).
3. At least 4 pairs of corresponding coordinates are needed to estimate the 12 parameters, i.e., 12 equations and 12 unknowns.
4. Affine transformations can be combined easily via matrix multiplication alone.

$$T \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

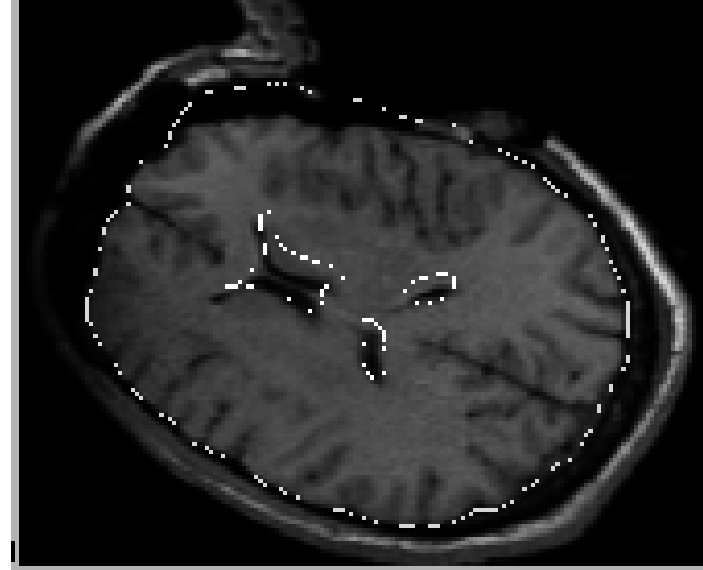
Affine transformation for 3D non-rigid registration

5. Derivations (in class)

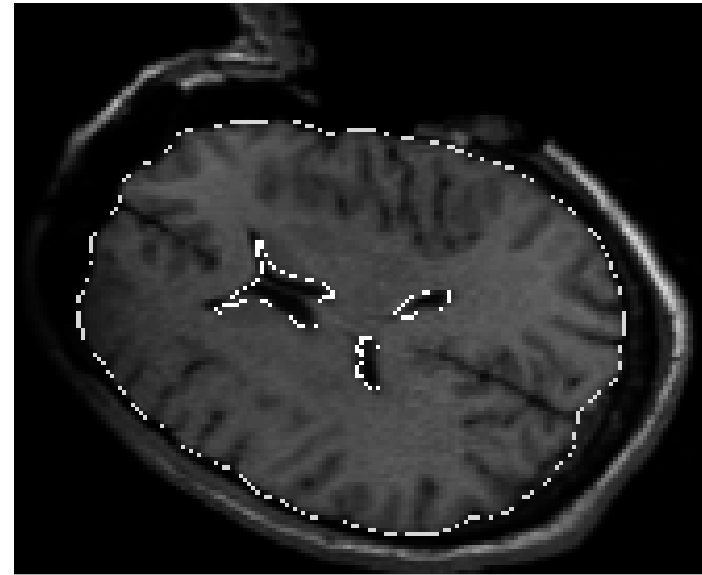
Example in Medical Image Analysis



Preoperative image



Intra-operative images



Affine transformation for 2D non-rigid registration

1. Shearing and scaling are included.
2. Beyond rigid body motion (translations and rotations only).
3. At least 3 pairs of corresponding coordinates are needed to estimate the 6 parameters, i.e., 6 equations and 6 unknowns.
4. Affine transformations can be combined easily via matrix multiplication alone.

$$T \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

Affine transformation for 2D non-rigid registration

5. Derivations (in class)

Objective functions

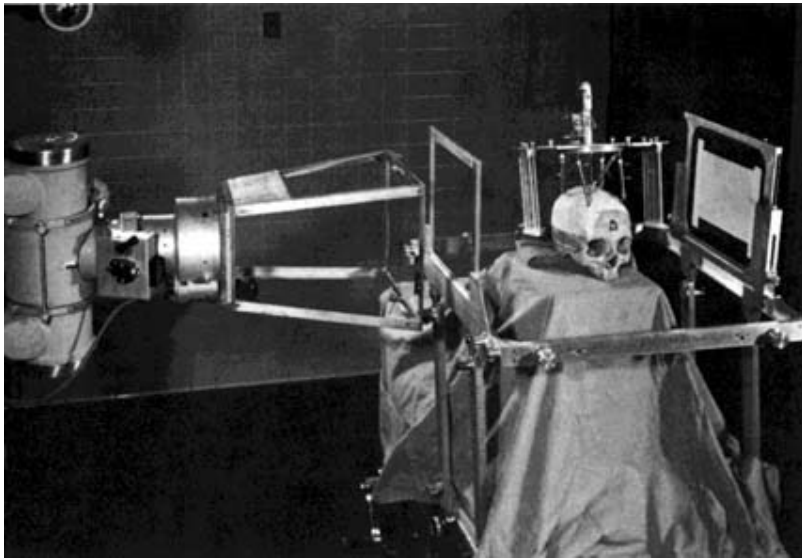
1. External markers
2. Internal markers (features)
3. Internal markers (voxel intensity)

Example: External markers



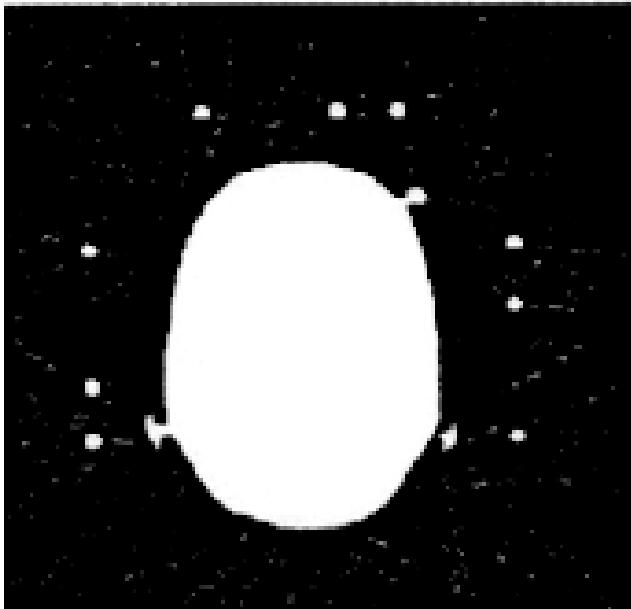
External markers

1. Stereotactic frame is rigidly attached to bone, usually the skull.
2. It is used to effect a registration between the physical patient and an image of that patient, typically for guiding biopsy needles, etc.

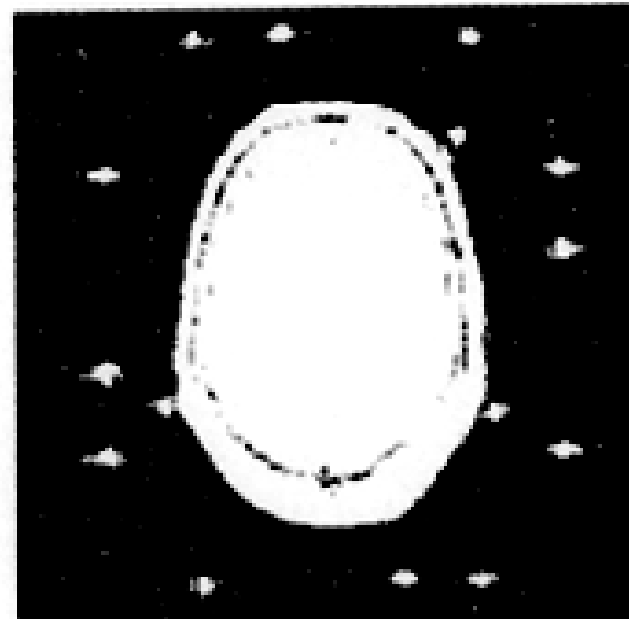


External markers

1. Fiducial markers are used and attached to a patient throughout the various acquisitions.
2. Point-based markers can be skull-implanted, skin-attached or bone implanted (less prone to consistent motion and more accurate)



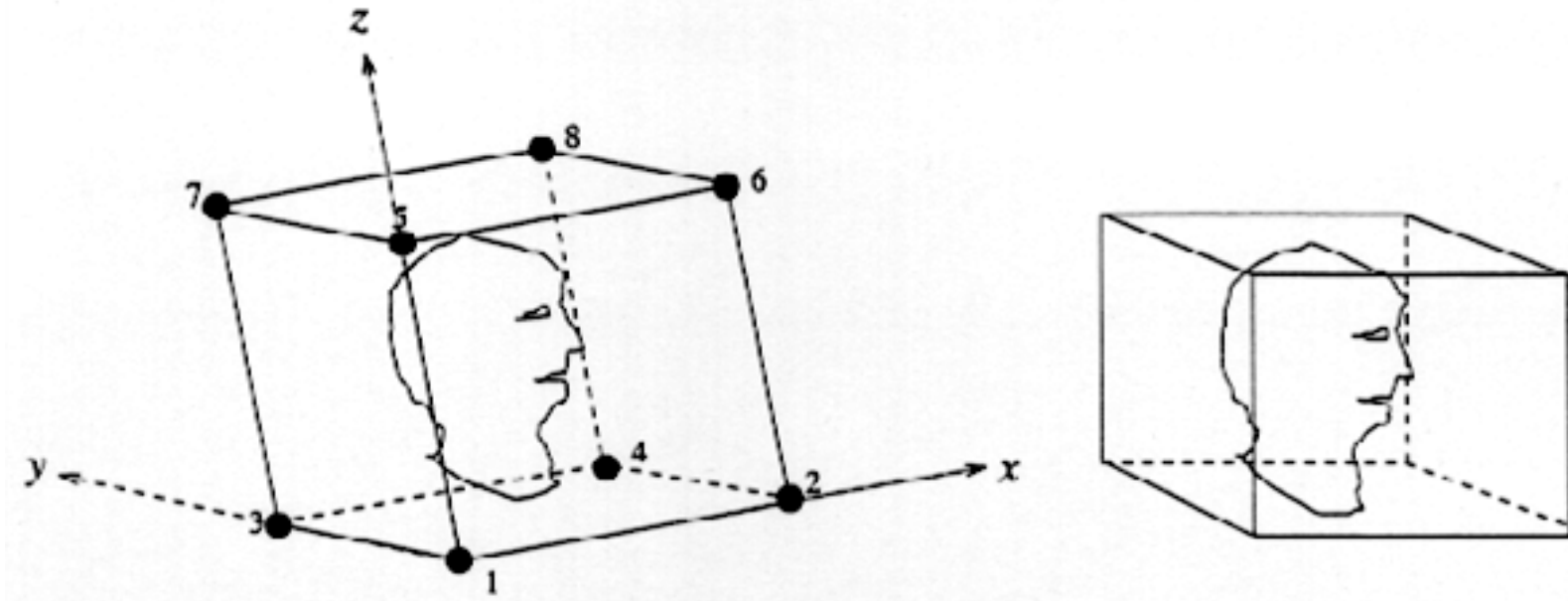
CT



MR

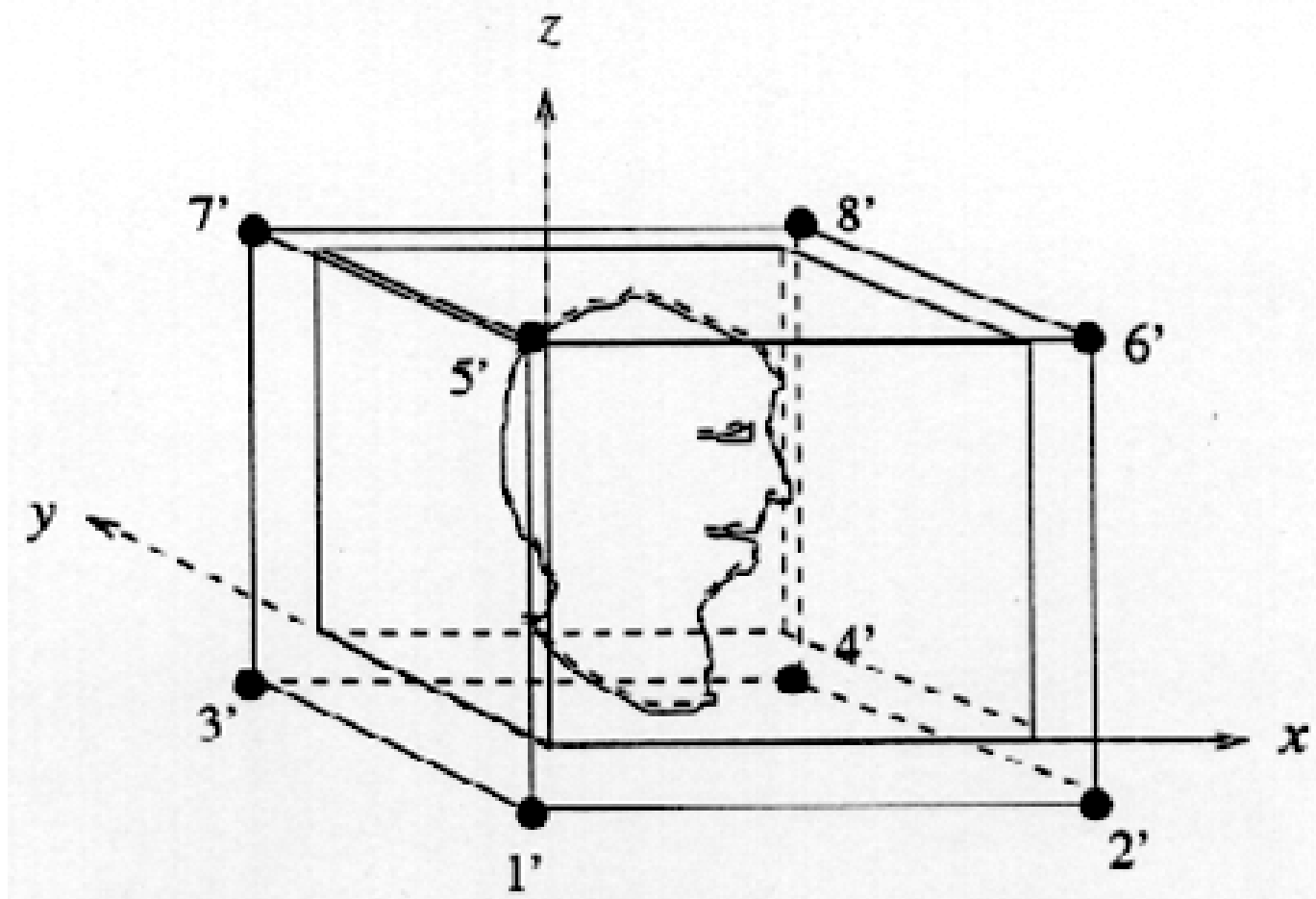
Measurement of registration errors

1. Markers can be seen and located in the images as tiny 3D volumes. Each marker is then represented by a point, which is defined as the volume centroid.



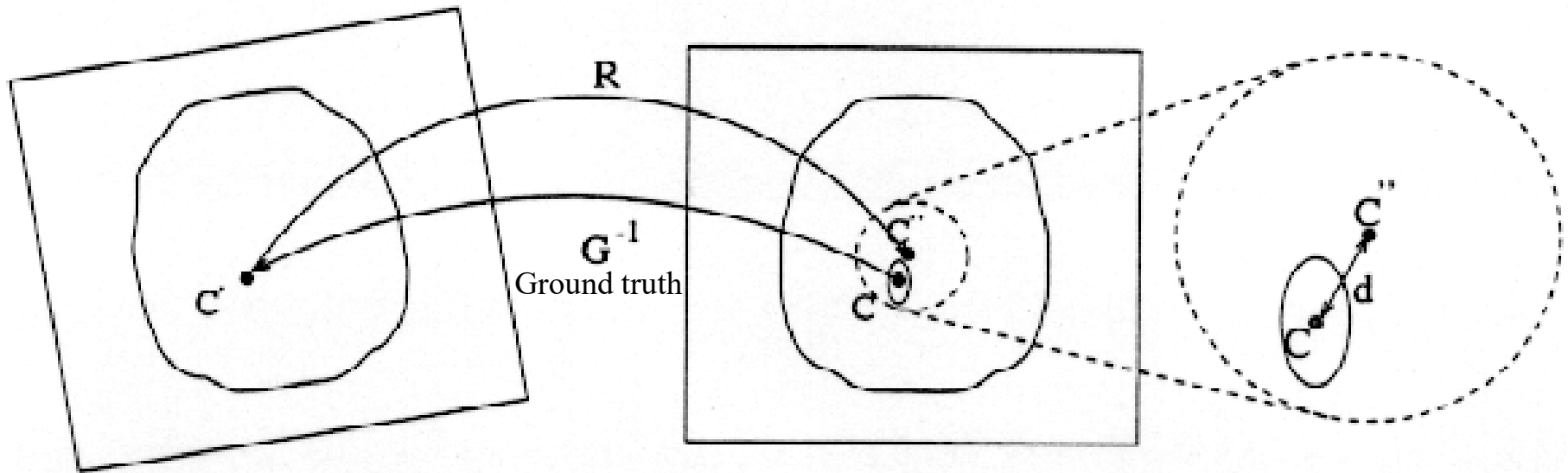
Measurement of registration errors

2. Each point is used to guide/evaluate the transformation.



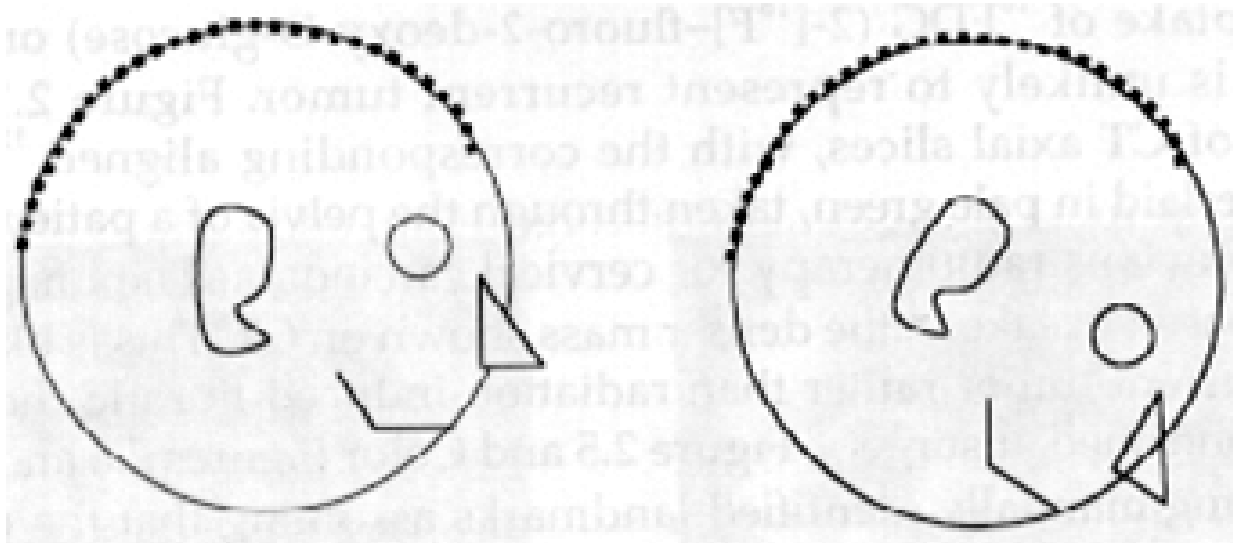
Measurement of registration errors

3. The registration error is measured by the Euclidean distance between the corresponding points in $u(\vec{x})$ and $v(\vec{x})$.



Internal markers: surface-based registration

1. The corresponding surface may be identified and used for registration rather than external markers. Need segmentation.
2. The corresponding surfaces are delineated in the two imaging modalities and are used to guide the transformation, which minimizes the distance (or some measure of distance) between the two surfaces.



Internal markers: edge-based registration

1. Image and Gaussian smoothing

An image $I(\vec{x})$ Smoothed image $I'(\vec{x}) = I \otimes G$

$\otimes$ denotes convolution

G denotes the zero-mean
Gaussian distribution

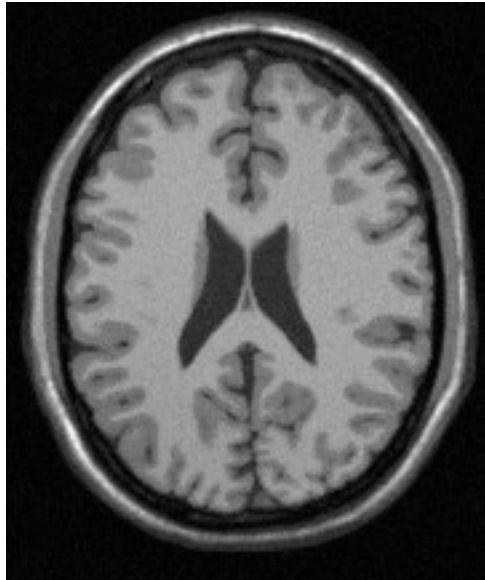
2. Gradient functions (magnitude of gradient)

$g(\vec{x}) = |\nabla I'(\vec{x})|$ Value of g increases as the gradient increases.

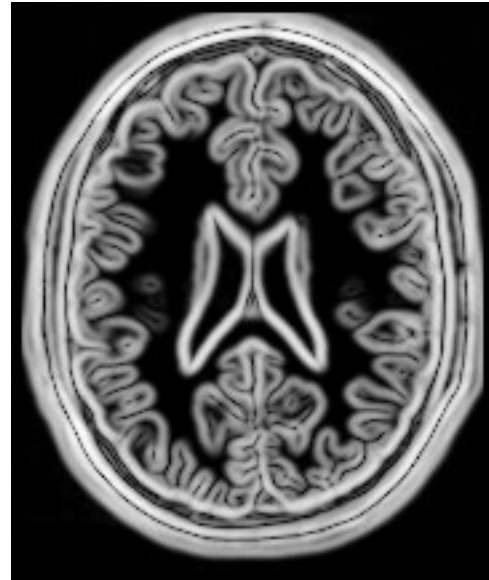
$g(\vec{x}) = \frac{\sigma^2}{\sigma^2 + |\nabla I'(\vec{x})|^n}$ Value of g decreases as the gradient increases.

Internal markers: edge-based registration

$I(\vec{x})$



$g(\vec{x})$



3. Edges are extracted/segmented based on their values of the gradient function.
4. Edges are then used to guide the transformation.

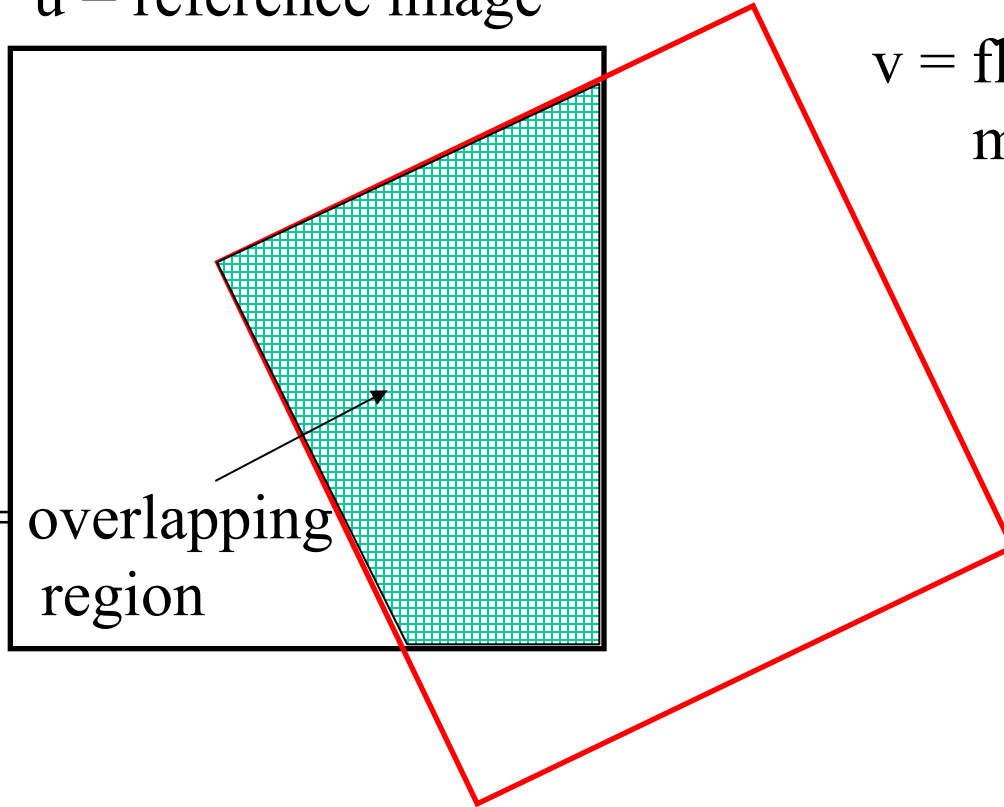
Internal markers (voxel/pixel intensity)

1. Sum of Squared Difference (SSD)
2. Sum of Absolute Difference (SAD)
3. Correlation Coefficient (CC)
4. Mutual Information (MI)

u = reference image

v = floating image/
moving image

D = overlapping
region



Sum of squared difference (SSD)

1. Sum of squared intensity difference (widely use)

$$\text{SSD} = \frac{1}{N} \sum_{\vec{x} \in D} \left[u(\vec{x}) - v(T(\vec{x})) \right]^2$$

, where D represents a domain in space, which can be the overlapping domain or the entire domain.

2. SSD is normalized so that it is invariant to the number of voxels in D.

Sum of absolute difference (SAD)

1. Although SSD is widely used, it is very sensitive to a small number of voxels that have very large intensity differences between u and v (outliners).

Large quadratic penalty.

2. SAD is defined as

$$\text{SAD} = \frac{1}{N} \sum_{\vec{x} \in D} \left| u(\vec{x}) - v(T(\vec{x})) \right|$$

, where D represents a domain in space, which can be the overlapping domain or the entire domain.

3. SAD can reduce the effect of these outliners.

Correlation coefficient (CC)

1. Correlation Coefficient

$$CC = \frac{\sum_{\vec{x} \in D} (u - \bar{u})(v - \bar{v})}{\{\sum_{\vec{x} \in D} (u - \bar{u})^2 \sum_{\vec{x} \in D} (v - \bar{v})^2\}^{\frac{1}{2}}}$$

where D represents a domain in space, which can be the overlapping domain or the entire domain, $\bar{u}$ and $\bar{v}$ represent means of u and v respectively.

2. CC assumes that there is a linear relationship between the intensity values in the images.
3. $-1 \leq CC \leq 1$

Mutual Information (MI)

1. Mutual information

$$\text{MI}(u, v) = h(u) + h(v) - h(u, v)$$

where $h(\cdot)$ represents the entropy of a random variable and $h(x, y)$ represents the joint entropy of two random variables.

More info: [https://en.wikipedia.org/wiki/Entropy_\(information_theory\)](https://en.wikipedia.org/wiki/Entropy_(information_theory))

$$h(x) = - \int p(x) \ln p(x) dx$$

$$h(x, y) = - \int p(x, y) \ln p(x, y) dx dy$$

2. (Information) Entropy is a measure of *uncertainty* / *randomness*.

<https://pymedix.com/2019/03/12/history-of-image-registration-part-3/>

https://matthew-brett.github.io/teaching/mutual_information.html

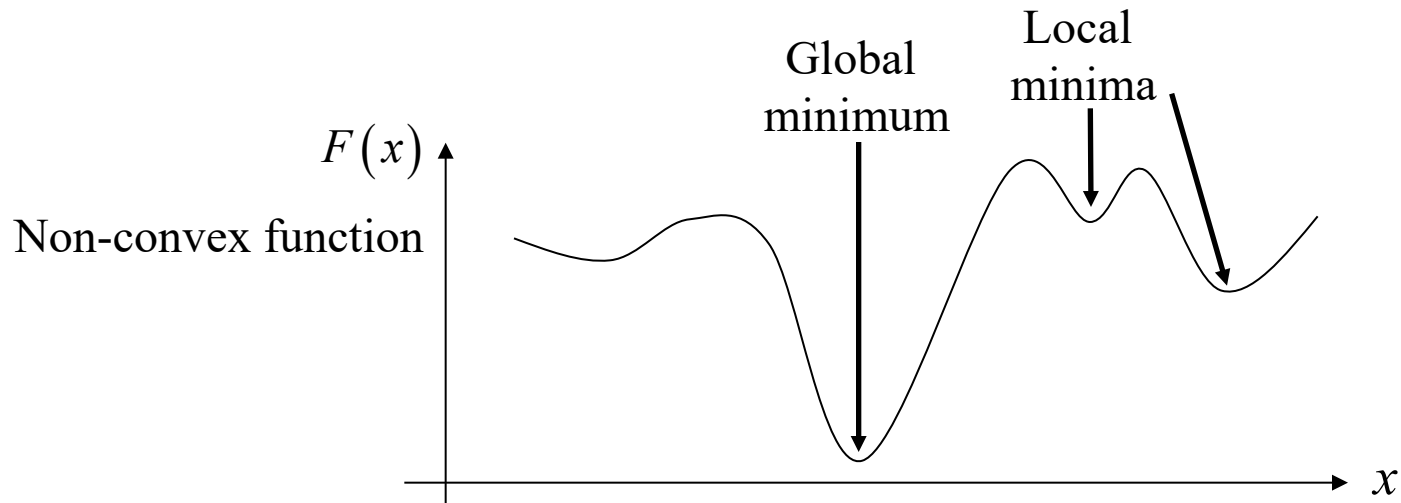
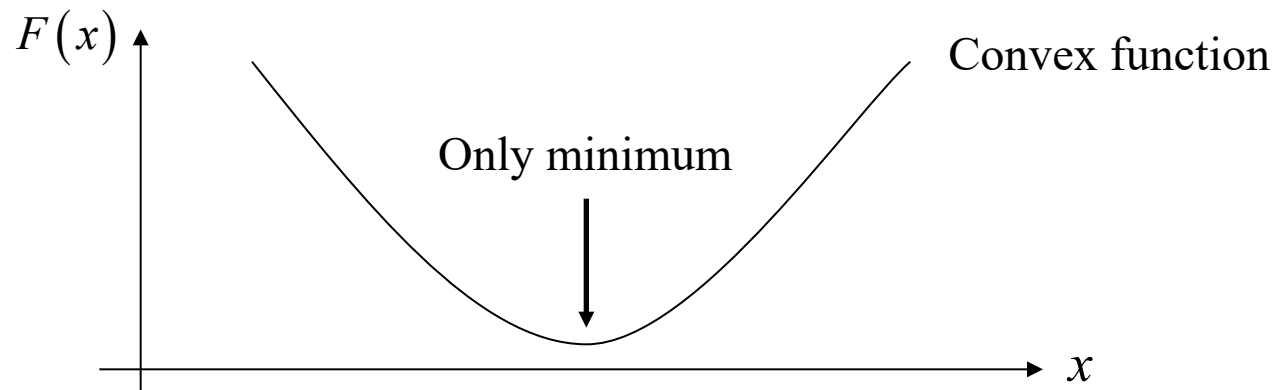
Mutual Information (MI)

3. The first term is the entropy of image u .
4. The second term is the entropy of image v which overlaps with u .
5. The third term is the joint entropy which encourages image matching.
6. The two images are matched when MI is maximized.

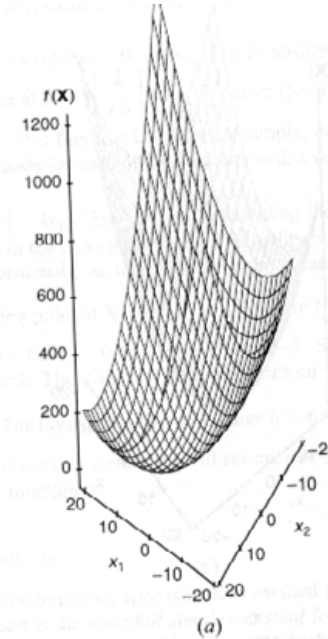
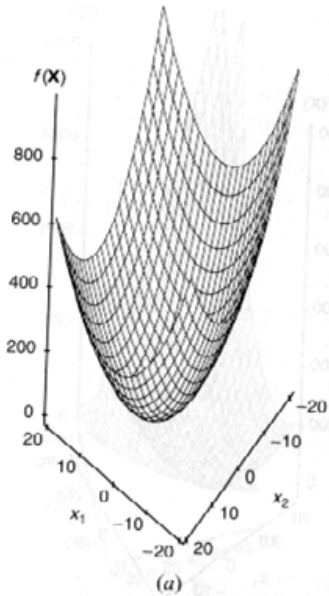
Optimization methods

Goal of optimisation

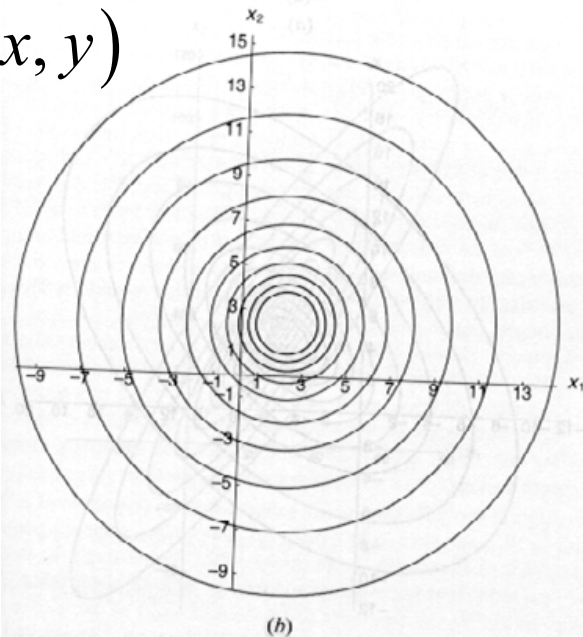
Find x that minimizes/maximizes $F(x)$, which is an objective function.



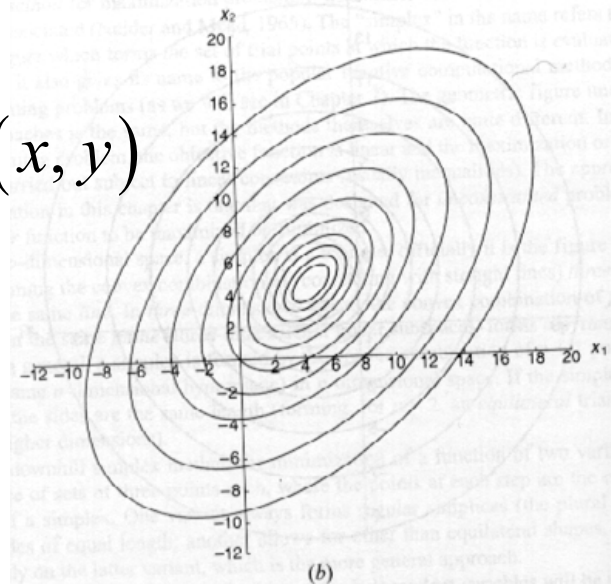
Goal of optimisation



$F(x, y)$



$F(x, y)$



Goal of optimisation

For 3D rigid body transformation problem, find $T = (\theta_x, \theta_y, \theta_z, t_x, t_y, t_z)$ that minimizes/maximizes the objective function, which is given as

$$F(u(\vec{x}), v(T(\vec{x})))$$

1. Need to search locally the optimal solution in 6-dimensional space.
2. Global exhaustive search is often too computationally expensive.

Powell's method

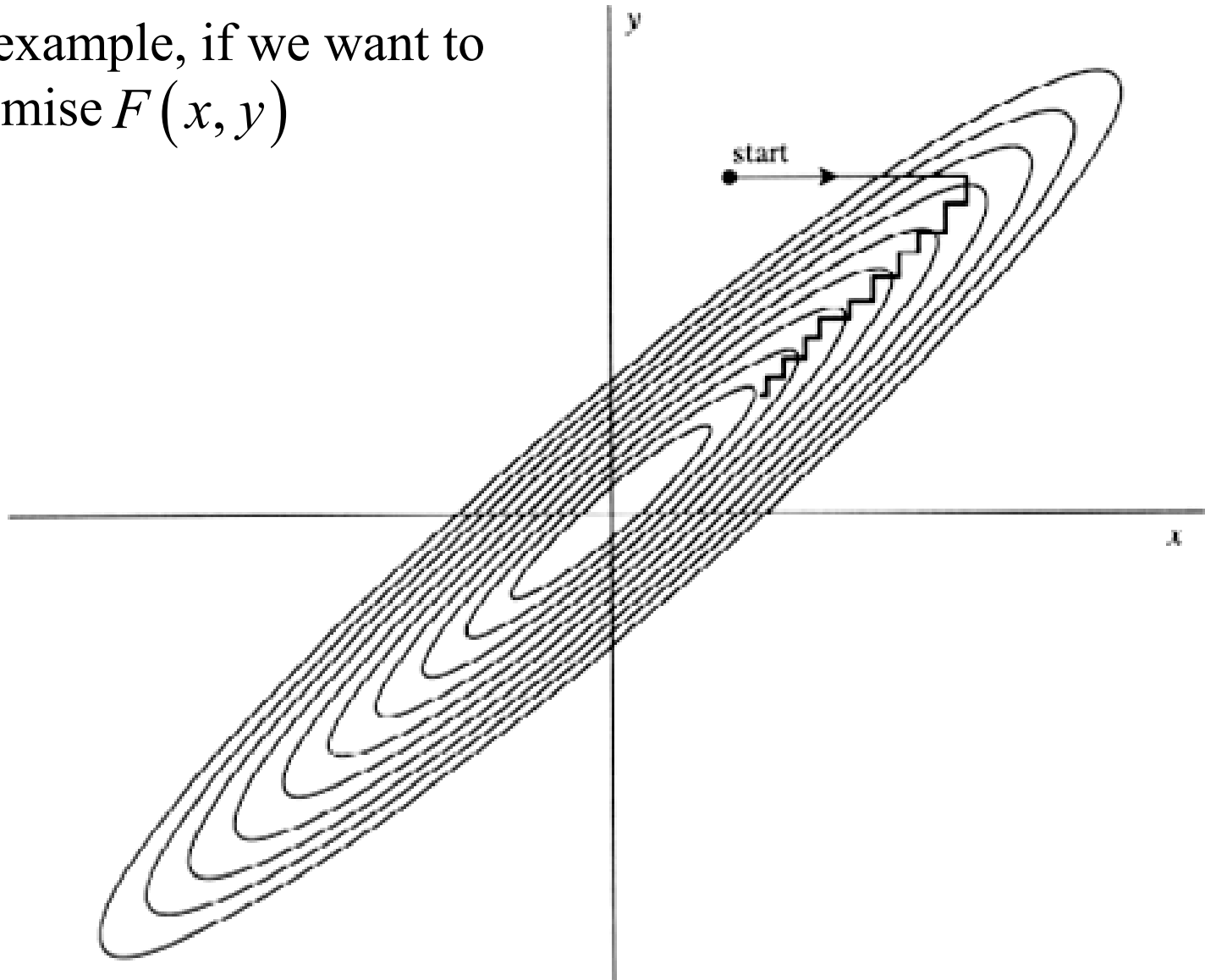
1. Assume that the multidimensional minimization methods consist of sequences of one-dimensional (1D) line minimization.
2. Definition of the 1D line minimization: Given as input the vectors $\vec{P}$ (any point) and $\hat{n}$ (direction vector), and the function F , find the scalar λ that minimizes $F(\vec{P} + \lambda\hat{n})$. Replace $\vec{P}$ by $\vec{P} + \lambda\hat{n}$. Done when the change in $\vec{P}$ is insignificantly small.
3. This method does not require calculations of the gradient. Computationally simple.
4. If the basis unit vectors, $e_1, \dots, e_6$, are taken as the directional vectors for 1D line minimization, then a set of directions is formed. Assume that the vectors are linearly independent.

Powell's method

5. The minimization procedure: minimize along the first directional vector, then along the second vector, ... repeat until the last directional vector. The cycle is repeated until the change in $\bar{P}$ is significantly small during the entire cycle.
- 6.
7. Faster if conjugate directions are used. If one moves along the direction $\hat{n}_1$, then the second direction is the direction that the gradient does not change its direction.

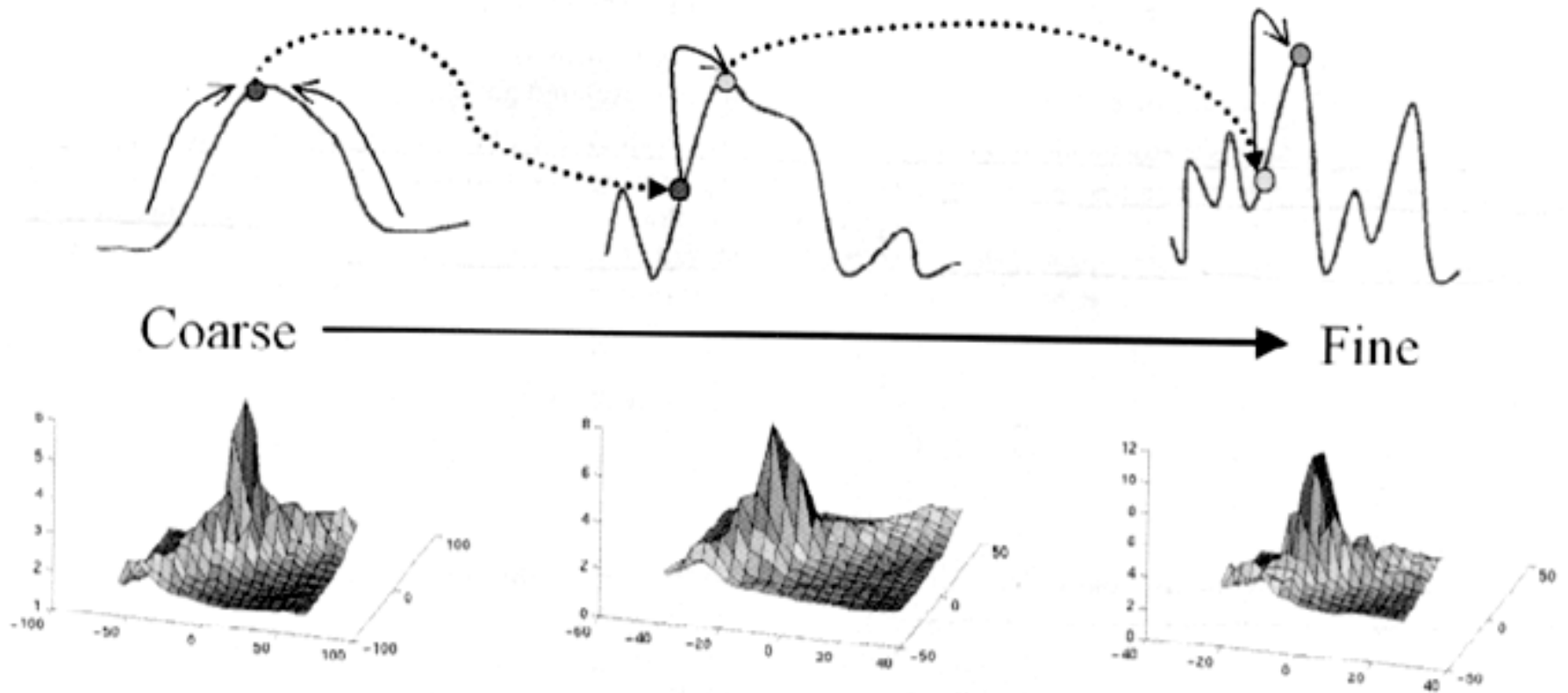
Powell's method

For example, if we want to
minimise $F(x, y)$



Search strategies: multi-resolution approach

1. Multi-resolution approach is also called coarse-to-fine approach.



Search strategies: multi-start approach

